Id: 1578465

Participatory Educational Research (PER)
Vol.12(5), pp. 23-42, September 2025
Available online at http://www.perjournal.com
ISSN: 2148-6123
http://dx.doi.org/10.17275/per.25.62.12.5

Factors contribute to developing digital competencies in economics education in Vietnamese universities: evidence from a lecturer survey

Tran Thi Thuy Ngoc*
Vietnam Youth Academy, Vietnam
ORCID: 0009-0000-8117-6101

Nguyen Nghi Thanh
National Academy of Public Administration, Vietnam
ORCID: 0000-0002-0791-8063

Article history

Received:
03.11.2024

Received in revised form:
18.02.2025

Accepted:
13.05.2025

Key words:
Digital Competencies; Economics Education; Vietnamese Universities; Curriculum Integration; Faculty Expertise; Institutional Support

With the rise of the digital economy being a salient feature of Vietnam's higher education sector, the pressure is on universities to provide their students with the digital competencies needed to enable academic achievement and future professional success. This study examined the antecedents of digital competencies in economics education by identifying four important areas: curriculum, faculty development and expertise, technology, and institutional supports and policies. Data were collected from 200 faculty members from six of Vietnam's leading economics universities and analysed through Structural Equation Modeling (SEM). The study found that while technological infrastructure is a necessary condition for digital competencies, it has a limited ability to explain when it is not localized within supportive frameworks of regulations and policies in their pedagogies. Moreover, we found a robust relationship between students' digital competencies and their ability to engage in academic life and apply economic concepts in real-world experiences. This suggests that digital competences are not only technical skills, but also fundamental enablers of valuable learning and being adaptable professionals. In conclusion, if universities are to provide economics education that contributes to the expectations stemming from a digital economy, it is essential to develop and sustain faculty development in curriculum redesign, establish supportive institutional policies, and use technology for design not coincidental purposes to reduce the likelihood of careless mistakes..

Literature review

Digital competencies in economics education

In the context of economics education, digital competencies refer to a set of integrated knowledge, skills, and attitudes that enable individuals to effectively use digital technologies for economic analysis, interpretation of data, problem-solving, and professional communication. These competencies are not limited to basic computer literacy but include advanced proficiencies such as operating econometric software, utilizing digital simulation

* Correspondency: tranthuyngoc19111974@gmail.com

tools, and engaging in digital research methods tailored to economic contexts. Numerous studies have emphasized the critical role these competencies play in preparing students for the demands of a digitized economic environment (Zhyvko & Petrukha, 2023).

At a fundamental level, digital competencies in economics require the ability to use specialized software for economic modeling and quantitative analysis, such as STATA, R, or EViews, which are increasingly embedded within modern economics curricula (Wickramasinghe et al., 2023). As Johnson and Lee (2021) explain, such tools allow students to perform complex statistical analyses, interpret large datasets, and challenge theoretical assumptions through empirical applications. They note that "the majority of economists have a better understanding of economic phenomena when they can recreate theoretical models in empirical environments and evaluate the validity of such models using real-world data."
Beyond technical proficiency, digitally competent students must also demonstrate information literacy—effectively sourcing credible economic data, navigating academic databases, and understanding digital ethics such as privacy and intellectual property laws (Salvi et al., 2021; Zvereva, 2023). Patel and Wang (2022), along with Rafi et al. (2019), highlight the growing importance of online research skills, particularly in using economic databases and repositories. Moreover, the literature points to the value of simulation-based learning and modeling tools, which not only strengthen theoretical comprehension but also foster applied skills essential for real-world problem-solving (Anderson & Kumar, 2019; Voskresenskaya et al., 2020).

Collaborative digital platforms also form a vital component of digital competencies by promoting teamwork, communication, and problem-solving in online environments—skills increasingly valued in the labor market (Gallagher et al., 2020; Jou et al., 2023). Furthermore, effective engagement with e-learning environments enables flexible, self-directed learning and prepares students to adapt to various digital learning modalities (Guo, 2023). Taken together, the literature affirms that digital competencies in economics education are multifaceted and evolving, requiring universities to incorporate both theoretical and applied digital skills into their curricula to cultivate graduates capable of thriving in a data-intensive, technology-driven economy.

Factors contribute to developing digital competencies in economics education

Curriculum Integration:

Scholarly work on the importance of curriculum alignment using digital tools and methods for rich digital competencies in the context of economics teaching has been well researched (Van Duy, 2023). One fundamental path for avoiding educational content becoming meaningless or inconsequential in the context of digital times is the intentional integration of digital resources and analytical software in the economics curriculum (Chen, 2022) Bessonova and Battalov (2020) highlight the need for updating the economics course for the use of contemporary digital tools and software, and consider the integration of both of these fundamental to bridging theoretical economic concepts learnt, and their operational activities derived learning under a technology-based umbrella. After which Kosovskihet et al. (2020) reinforced this notion with compelling evidence of the impact of integrating digital resources, which positively impacted students' analytical and problem-solving skills with economics course.

In addition, it is essential for curricular guidance to incorporate teaching approaches that

emphasize digital literacy and digital competence as an means increase student digital capacity. According to Sarva et al. (2023), the incorporation of curriculum has benefits on digital literacy, such as greater proficiency in software and deeper perspectives of digital data analysis in economic learning. Supporting this, Instefjord and Munthe (2016) note that students who were provided with learning tasks that included digital tools demonstrated higher levels of understanding of complex hypotheses due, in part, to the interactive and immersive features of the digital tool used in their learning. . Accordingly, this leads to the following research hypothesis:
Hypothesis 1 (H1): Curriculum integration has a positive and meaningful impact on digital competencies in economics education

Faculty Expertise and Development:

The importance of faculty expertise and development in promoting digital capabilities in economics education is a key theme in the current literature. It is widely understood that educators must be able and willing to use and teach with digital tools within their courses. For example, Thompson and Lee (2018) and Gupta et al. (2019) discussed educators' urgent need for the understanding and abilities to select and use digital technologies. They reference to technological literacy and not just technological literacy, but the understanding of how to pedagogically apply these technologies in their discipline for better economies learning outcomes.

Equally important is to make sure training and resources are available, and train faculty on advances in digital technologies and teaching practices, to support faculty use of the latest digital technologies. For example, Núñez-Canal et al. (2022) found teacher professional development programs in faculty development supported educators to be more digitally capable and resulted in better teachers in economics as they were incorporating new digital environments into their practices. More importantly, this study provided evidence that educator personal advancements of proficiency with digital tools produced more academically consistent applications of digital tools into practice, which resulted in economics teacher-student interactions that were more enriching and effective.

The implications of these publications point to an implicit responsibility that teachers will "model" effective online teaching and learning. In the literature, there is a recommendation that a priority for faculty development online programs should be on-going and absolutely essential in economics and/or sociology, because the ability of faculty to teach and effectively use digital environments is essential to the quality of students entering the world of data and technology. The scholarly line does hold that faculty development is essential for building the capacity to produce an economics education of the highest quality in the digital age. Accordingly, this leads to the following research hypothesis:
Hypothesis 2 (H2): Faculty expertise and digital development positively and meaningfully impact digital competencies in economics education.

Technological Infrastructure:

Educational research has highlighted the significance of technological infrastructure for the development of digital capabilities in students of an Economics program. Various resources provided by Technological Infrastructure, such as high-speed internet, digital libraries, and specialized software, have been classified as indispensable learning-related resources for students in a digital context (Ngoasong, 2018; Merriam & Baumgartner, 2020). In terms of immediacy, Warschauer and Matuchniak (2010) explicitly note that high-quality

technological resources are contextually relevant, to signify how any technological infrastructure that students receive in the Economic class is imperative for developing and maintaining digital capabilities relevant in the economics context. These scholars demonstrate their assertion whether through theoretical or empirical means that technological infrastructure is viewed as a lynchpin in educational construction, having even argued establishing that it directly enhances economic learning outcomes.

Corrall (2012) addresses the immeasurable value of digital libraries as abundant resources of knowledge that are instrumental in the processes of high-level economic research and analysis. Digital spaces furnish a vast range of severely relevant information and materials that are crucial to the diligent undertaking of economic analysis and inquiry. In a closely related manner, Rojewski (2002) delves into the specific role that specialized software plays in the training of future economists. The author argues that working with the tools they will use in the profession provides students with what they really need to ensure their education is equipping them for the multifaceted responsibilities of the profession.

Altogether the ideas gleaned from these writings demonstrate the utter and total importance of the technology infrastructure in developing digital literacy by students of economics. In the absence of the firm and foundational support of robust and trustworthy technology infrastructure, the practical processes of developing those digital competencies required for the economically situated processes of students operating in a global context is in peril. Hence, we propose the following research hypothesis:
Hypothesis 3 (H3): Technological infrastructure has a positive and meaningful impact on digital competencies in economics education

Support and Policies:

The increasing significance of both institutional strategies and institutional support to the development of digital competencies, and hence digital learning, in higher education are important areas of educational research. In the case of economics education, the university or college provides the structure and the policies for the effective integration of digital tools and digital resources into the economics curricula. The structure and policies are an important form of support, helping ensure that the student experiences obtain the digital competencies to be effective in working in the modern economy (Jones, 2019). For example, evidence from a seminal study, (Hennessy, et al, 2015), presented findings on the critical role of institutional policies in shaping the effective integration of students' educational experiences using technology in economics. The researchers found programs located in institutions with supportive technological and pedagogical arrangements were much more likely to utilize digital resources and thus make digital resources a part of their economics education.

Similarly, Lim & Morris (2009) noted that programs with institutional efforts that resulted in training on how to use digital tools, or to have digital tools available, are critical in ensuring the conditions for effective digital learning would likely occur. They found participants who were trained to use technology-enabled learning in economics and have the specific course-related skills were much more able, or proficient, in operating the technologies in a manner that permitted digital analyses of economic performance, in ways and with other outcomes much better than students taking the same course in a more traditional, contemporary learning environment.

Correspondingly, Greenhow and Lewin's (2016) study indicates that institutional support through policy initiatives is critical to both shifting the culture of the university classroom and

continuing the culture of digital innovation and learning. They express that when there is a solid institutional commitment to this type of mode of excellence through digital means, it "will increase the learning that is undertaken so much that students will be in a place where their quality of learning is as good as experience can teach them." This can enable students to participate within an economy for the first time in a real way.

These studies demonstrate how institutional support and policy development are significant catalysts for developing digital competencies in Economics education. Even more, these authors build a strong case for the university let us say, to unambiguously develop frameworks or policies that are institutionally supportive of the entry of digital technologies and digital resources into the learning space. It is conceivable that this strand of the literature holds one of the keys to the ultimate improvement of Economics education relevance and quality to the moral responsibilities of life within the digital age. Hence, we propose the following research hypothesis:
Hypothesis 4 (H4): Institutional support and policies have a positive and meaningful impact on digital competencies in economics education

Institutional Support and Policies:

Investigating the role digital competencies play in economics education in terms of student engagement and practice is important in studying the transformative aspects of digital competency in economics education. Although few studies have investigated student engagement in economics education, the field recognizes the importance of digital competencies.

Ifenthaler (2012) conducted a definitive study that demonstrated digital competencies functioned to engage economics students. Ifenthaler concluded that the more students were experienced using digital competencies, the higher the student engagement. Economically analyzing data using digital competencies, modeling economic phenomena, and solving complex economic problems drew in student engagement. Students were engaged in economics through digital competencies, which allowed for the use of simulations, data analysis projects, and online collaborative workspaces, accurately mentioned by Laurillard (2013). These were critical in that they afforded opportunities for students to understand economic theories and concepts in practice. It is acknowledged that advanced digital competencies were a prerequisite for students moving from the theoretical learning of economics to the practical use of understanding in an educational setting.

The significance of digital competencies for student collaboration in learning more generally has been a key element of Selwyn's (2014) scholarship, and as such, there is also an aspect of collaborating on projects, sharing and pooling economic insights, and developing responses to an economic problem. Digital competencies supported students being collaborative economies learners and as such represented a meaningful development for richer and more sophisticated forms of learning economics and closer to the type of teamwork and solutions found in the day-to-day work of the economist.

In summary, digital competencies, improve student engagement and practical work in economics education, through the collaborative nature of work, greater interactivity, and enabling students to enact the theories and concepts of economics in real-world practical settings. Therefore, we propose the following research hypothesis:
Hypothesis 5 (H5): Digital competencies in economics education have a positive and meaningful impact on Support and Policies

Based on the research hypotheses, the following research model is proposed (Figure 1):

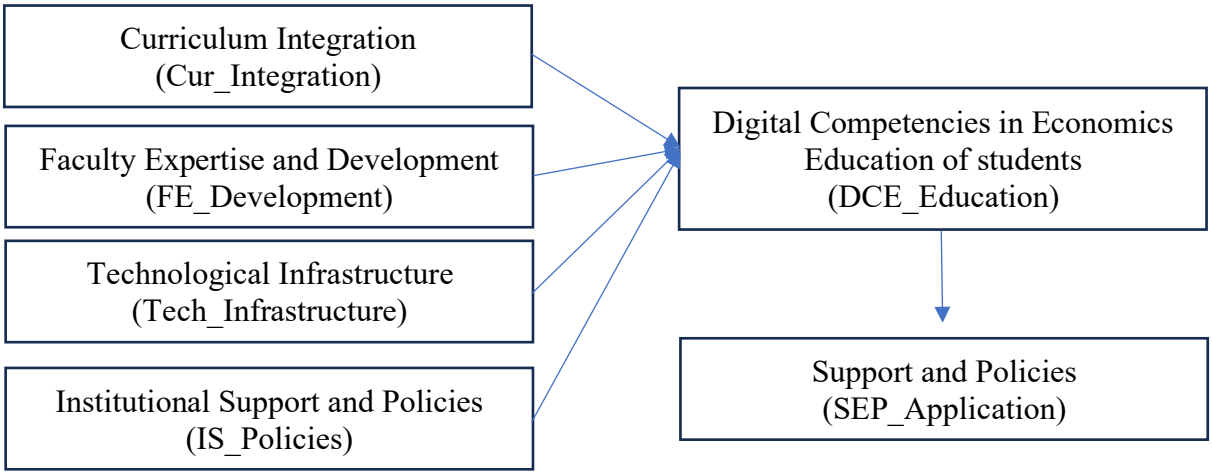


Figure 1. Proposed research model

Methodology

Instrument

The data collection instrument utilized in this study was a comprehensive questionnaire, originally developed from a review of related literature and consultation from two experienced educational researchers (Oosterveld, 1996). The questionnaire consisted of two main sections: section one consisted of questions about the characteristics of the respondents, and section two comprised the substantive area of inquiry of this research study (Martin, 2006). In the development of the questionnaire instrument, a two-step validation and refinement process were implemented to ensure both the relevance and reliability of the measures used to assess digital competencies in economics education. Step one involved the administration of a pilot survey to 40 economics lecturers across three universities. Respondents were asked to complete the draft version of the questionnaire and provide written feedback on the clarity, relevance, and phrasing of each item. Based on the pilot data and qualitative feedback, several adjustments were made. These included rewording ambiguous statements, reducing item redundancy, and refining the Likert scale descriptors to enhance interpretability. For example, one item that initially asked about "technological skill application" was revised to specifically mention "the use of digital simulation software in teaching economic concepts," aligning more closely with the study's focus.

Step two involved consultation with two domain experts in economics education and educational assessment. The experts reviewed the updated questionnaire and suggested further refinements, particularly in aligning the language with academic and pedagogical terminology. Their feedback led to improved construct alignment and better thematic grouping of items into dimensions such as curriculum integration, faculty development, technological infrastructure, institutional support, and student engagement. The final version of the instrument included 24 items distributed across six constructs, each measured with four items on a five-point Likert scale ranging from 1 ("Strongly Disagree") to 5 ("Strongly Agree").

Although the manuscript originally referred to an appendix, the detailed list of survey items was inadvertently omitted in the submitted file. The revised manuscript will include the full

list of survey questions in the Appendix, categorized by construct, to enhance transparency and replicability.

Participants

The researchers randomly selected participants, all of whom were full-time lecturers at six leading economics-focused universities in Vietnam: Foreign Trade University (FTU), Ho Chi Minh City University of Economics (UEH), National Economics University (NEU), The University of Danang – University of Economics (DUE), Thuong Mai University (TU), and University of Economics – Hue University (HUE). These institutions were selected based on three key criteria.

First, institutional reputation: Each university has been consistently ranked among the top economics and business education institutions in Vietnam, with strong national influence and regional leadership in curriculum innovation, faculty research output, and partnerships with government and industry. Second, curricular engagement with digital transformation: These universities are known for their early adoption of digital tools and platforms in economics education, making them particularly relevant contexts for examining digital competencies in teaching and learning. Third, geographical and demographic diversity: The selection includes institutions from both northern, central, and southern Vietnam, which allows for a broader representation of institutional conditions, faculty development programs, and infrastructural environments.

The inclusion of these universities was thus intended not only to capture excellence but also to reflect a cross-section of the economic education landscape in Vietnam, enhancing the external validity and applicability of the study's findings across different institutional settings. The study participants were selected based on the significance of their institution towards achieving the study objectives. The sampling method employed quota-sampling of 200 lecturers to ensure the representativeness of the research results and conclusions vis-a-vis the target population (Etikan & Bala, 2017). The research team directly gave the survey to participants and also directly collected the survey instrument once the participants provided their responses using pencils.

The researchers directly administered printed questionnaires to participants during faculty meetings and academic seminars at the selected institutions. This in-person distribution approach enabled the research team to provide immediate clarification on any uncertainties, increasing participant understanding and engagement. Furthermore, respondents were given time during the sessions to complete the survey and return it on-site. This on-the-spot completion method, combined with institutional support from faculty leadership and the assurance of anonymity, contributed to the exceptionally high response rate.

Indeed, a 100% response rate was achieved (N = 200), which, while uncommon in social science research, was made possible by this controlled, facilitated environment and the use of follow-up reminders prior to the in-person sessions. Although incentives were not offered, the strong academic interest in digital transformation and professional relevance of the research topic likely also contributed to the high participation rate. To mitigate non-response bias, the researchers ensured that the sample included a balanced distribution of faculty by age, academic rank, and university, as detailed in Table 1. These measures helped ensure that the data accurately reflect the broader lecturer population at Vietnam's leading economics universities.

Ethical Considerations

Central to this research was an emphasis on ethical integrity. Participants were fully informed the research is taking place and ensured that their responses are confidential and anonymous (Macfarlane, 2010). On returning completed questionnaires, all participants were informed of consent, which indicates they understood they have rights and are free to withdraw from the research without question and to follow up with the researcher, if desired (Sikes & Piper, 2010). All data were collected and managed with the utmost principles of privacy and confidentiality; so that only biographical background and participants' responses to prompted questionnaire items can be reported here.

Table 1. Demographic characteristics of survey participants

		Major					
		Data Science and Economics		E-Commerce		Financial Technology	
		Count	Row N %	Count	Row N %	Count	Row N %
Age	Over 45 years old	7	23.3%	10	33.3%	13	43.3%
	31 35 years old	10	20.4%	16	32.7%	23	46.9%
	36 40 years old	18	19.6%	37	40.2%	37	40.2%
	40 45 years old	9	31.0%	9	31.0%	11	37.9%
Gender	female	18	23.1%	31	39.7%	29	37.2%
	male	26	21.3%	41	33.6%	55	45.1%
University	Foreign trade university	4	9.3%	22	51.2%	17	39.5%
	Ho Chi Minh City University of Economics	15	30.0%	16	32.0%	19	38.0%
	National Economics University	9	29.0%	10	32.3%	12	38.7%
	The University Of Danang/University Of Economics	9	32.1%	7	25.0%	12	42.9%
	Thuong Mai University	5	16.7%	12	40.0%	13	43.3%
	University of Economics/Hue University	2	11.1%	5	27.8%	11	61.1%
Education	Master	12	15.4%	27	34.6%	39	50.0%
	PhD	29	27.4%	38	35.8%	39	36.8%
	Postdoc	3	18.8%	7	43.8%	6	37.5%
Experience	1 5 years	13	28.9%	17	37.8%	15	33.3%
	10 15 years	13	19.7%	23	34.8%	30	45.5%
	6 10 years	6	15.4%	13	33.3%	20	51.3%
	more than 15 years	12	24.0%	19	38.0%	19	38.0%

Results and Discussion

Results

Reliability analysis

A measure of internal consistency, Cronbach's alpha assesses the degree to which items of a survey reflect a single construct, or latent trait (Vaske et al., 2017). This measure is most appropriate when multiple approaches or facets will be used to assess a single construct, with each of those facets influencing a solitary dimension. Values above 0.7 indicate strong internal consistency, while values above 0.6 are acceptable for exploratory purposes (Bernardi, 1994). This measure is most useful when working with continuous data from items written to assess a single construct.

Composite Reliability (CR) captures the reliability of a latent construct under Structural Equation Modeling (SEM) and provides a richer estimation of internal consistency compared to Cronbach's alpha, provided equal loading assumptions are not invoked (Bacon et al., 1995). CR is useful when determining reliability estimates for constructs that are operationalized through multiple indicators and a CR value above 0.7 suggests the latent construct has adequate reliability. Furthermore, it is essential when constructs are operationalized within SEM using multiple observed variables (Raykov, 1997).

Average Variance Extracted (AVE) reflects the amount of variance explained by the latent construct from the items, and is an additional measure of construct validity subsequent to the use of SEM. AVE is appropriate when the construct is in turn operationalized through multiple observed variables and an acceptable threshold for AVE is typically above 0.5, representing adequate convergent validity which indicates the latent construct is describing the variance in the determinant of the latent construct. There are scenarios in which lower AVE values may be considered, dependent on the nature of understanding how much variance in the construct can be explained.

Table 2. Summary of Reliability

Scales	Number of variables observed	Reliability coefficients (Cronbach Alpha)	Composite Reliability (CR)	Average Variance Extracted (AVE)
SEP_Application	4	0.765	0.767	0.453
IS_Policies	4	0.801	0.804	0.506
Tech_Infrastion	4	0.795	0.794	0.492
DCE_Education	4	0.756	0.757	0.439
FE_Development	4	0.777	0.776	0.467
Cur_Integration	4	0.779	0.780	0.471

The findings of the reliability and validity assessments for the research constructs are summarized in Table 2. All constructs demonstrated strong internal consistency, with Cronbach's alpha coefficients above the accepted threshold of 0.7, and Composite Reliability (CR) values exceeding 0.7 as well—indicating adequate construct reliability (Bacon et al., 1995; Vaske et al., 2017).

However, the Average Variance Extracted (AVE) values for four of the six constructs—SEP_Application (0.453), Tech_Infrastion (0.492), DCE_Education (0.439), and FE_Development (0.467)—fell slightly below the conventional benchmark of 0.5, which is

typically used to assess convergent validity (Fornell & Larcker, 1981). While this may initially raise concerns, it is important to note that AVE values marginally below 0.5 can still be acceptable, particularly when Composite Reliability exceeds 0.6, as is the case here. Scholars such as Malhotra (2020) argue that CR can serve as a more robust indicator of construct reliability in Structural Equation Modeling (SEM), and that lower AVE values do not necessarily undermine validity when supported by strong factor loadings and internal consistency.

In this study, all items exhibited factor loadings above 0.6, and each construct was supported by a conceptually coherent set of indicators, enhancing the interpretability of the constructs despite modest AVE scores. Moreover, the KMO value (0.906) and Bartlett's test of sphericity ($p < 0.001$) further support the appropriateness of the data for factor analysis. These results collectively indicate that the measurement model is sufficiently reliable and valid for continued SEM analysis.

Factor analysis

Exploratory Factor Analysis (EFA) is a statistical method utilized to assess latent constructs from measured variables and is applied in the earlier stages of research as constructs exist in the study site, but empirical and quantitative validation have not yet occurred (Fabrigar & Wegener, 2011). EFA is considered to be appropriate when data have a sufficient sample size relatively large, e.g., 5-10 observations per variable, as well as sufficiency in variable correlations, and such correlations indicate the possibility of some meaningful latent factors that relate measured variables. Exploratory Factor Analysis (EFA) was performed using Principal Component Analysis with Varimax rotation. The KMO value was 0.906, and Bartlett's test was significant ($\chi^2 = 1826.196$, $p < 0.001$), confirming sampling adequacy and factorability. Six factors with eigenvalues greater than 1 were extracted, explaining 62.06% of the total variance, which meets acceptable standards for social science research (Tabachnick & Fidell, 2013; Cattell, 1966; Ibikunle et al., 2021).

Table 3 Result of factor analysis

Rotated Component Matrix

	Component					
	1	2	3	4	5	6
IS_Policies3	.775					
IS_Policies2	.741					
IS_Policies1	.688					
IS_Policies4	.683					
Tech_Infrastructure4		.748				
Tech_Infrastructure3		.712				
Tech_Infrastructure2		.702				
Tech_Infrastructure1		.697				
DCE_Education1			.711			
DCE_Education3			.685			
DCE_Education2			.672			
DCE_Education4			.620			
FE_Development2				.727		
FE_Development3				.712		

FE_Development4	.699		
FE_Development1	.578		
SEP_Application1		.723	
SEP_Application2		.684	
SEP_Application3		.635	
SEP_Application4		.617	
Cur_Integration1			.774
Cur_Integration3			.685
Cur_Integration2			.679
Cur_Integration4			.641
Extraction Method: Principal Component Analysis. Rotation Method: Varimax with Kaiser Normalization.			
Rotation converged in 6 iterations. Kaiser-Meyer-Olkin Measure of Sampling Adequacy (KMO) = 0.906 Bartlett's Test of Sphericity (Chi-Square = 1826.196; df=276 ; sig.= 0.000) Extraction Sums of Squared Loadings = 62.059; Initial Eigenvalues = 1.063			

The results of the factor analysis to validate the research questionnaire are in Table 3. The results demonstrated that Bartlett's test of sphericity was significant (Sig. = 0.000) and KMO = 0.906 (> 0.5), confirming the observed variables are correlated in the population and are appropriate for factor analysis (Fabrigar & Wegener, 2011). The results of the factor analysis in Table 3 show that all variables have a factor loading coefficients = > 0.5 confirming that factor analysis is appropriate (Fabrigar & Wegener, 2011). The total load squared extraction of the six factors = 62.059 % (>50 %), confirming that the factor load square extraction of the extracted factors are accounting for significant amount of variance in the data. The initial eigenvalue of the six factors = 1.063 (> 1.00) confirming that the extracted factors had eigenvalues >= 1 and therefore were valid. These results provide evidence that the initial proposed research model was appropriate and valid for exploratory factor analysis.

SEM analysis

Structural equation modeling (SEM) is a sophisticated statistical method for testing complex research hypotheses related to observable and non-observable variables (dos Santos & Cirillo, 2023). SEM is useful for evaluating theoretical models and for examining direct and indirect relationships between constructs. There are specific goodness of fit indices for the model to be considered acceptable, including the Comparative Fit Index (CFI) and the Root Mean Square Error of Approximation (RMSEA). For an acceptable fit of the model, CFI values should be greater than 0.9 (a good fit), and RMSEA values should be equal to or less than .08 (indicating reasonable error of approximation in the population) (Tennant & Pallant, 2012). These indices ensure that the model demonstrates adequate accuracy in representing the actual social world, providing evidence SEM can be used to test hypotheses and demonstrate the validity of theoretical constructs in both social science and education research.

Table 4 Regression Weights

			Estimate	S.E.	C.R.	P	Decision
DCE_Education.	<---	Cur_Integration.	.210	.090	2.344	.019	H1 accepted
DCE_Education.	<---	FE_Development.	.469	.133	3.529	***	H2 accepted
DCE_Education.	<---	Tech_Infrastion.	.108	.098	1.104	.270	H3 not accepted
DCE_Education.	<---	IS_Policies.	.253	.081	3.114	.002	H4 accepted
SEP_Application.	<---	DCE_Education.	.902	.132	6.815	***	H5 accepted

Note: *** indicates $p < 0.001$

The analysis results, as shown in Figure 2, indicate that the SEM model satisfies the standard requirements. The Chi-square statistic = 272.794, GIF = 0.898 (approximately equal to 0.9), TLI = 0.978>0.90, and RMSEA =0.026<0.08. The results of the SEM analysis are presented in Table 4, which shows the relationship between the variables. Overall, the results suggest that the SEM model fits well with the data and represents the proposed research model well.

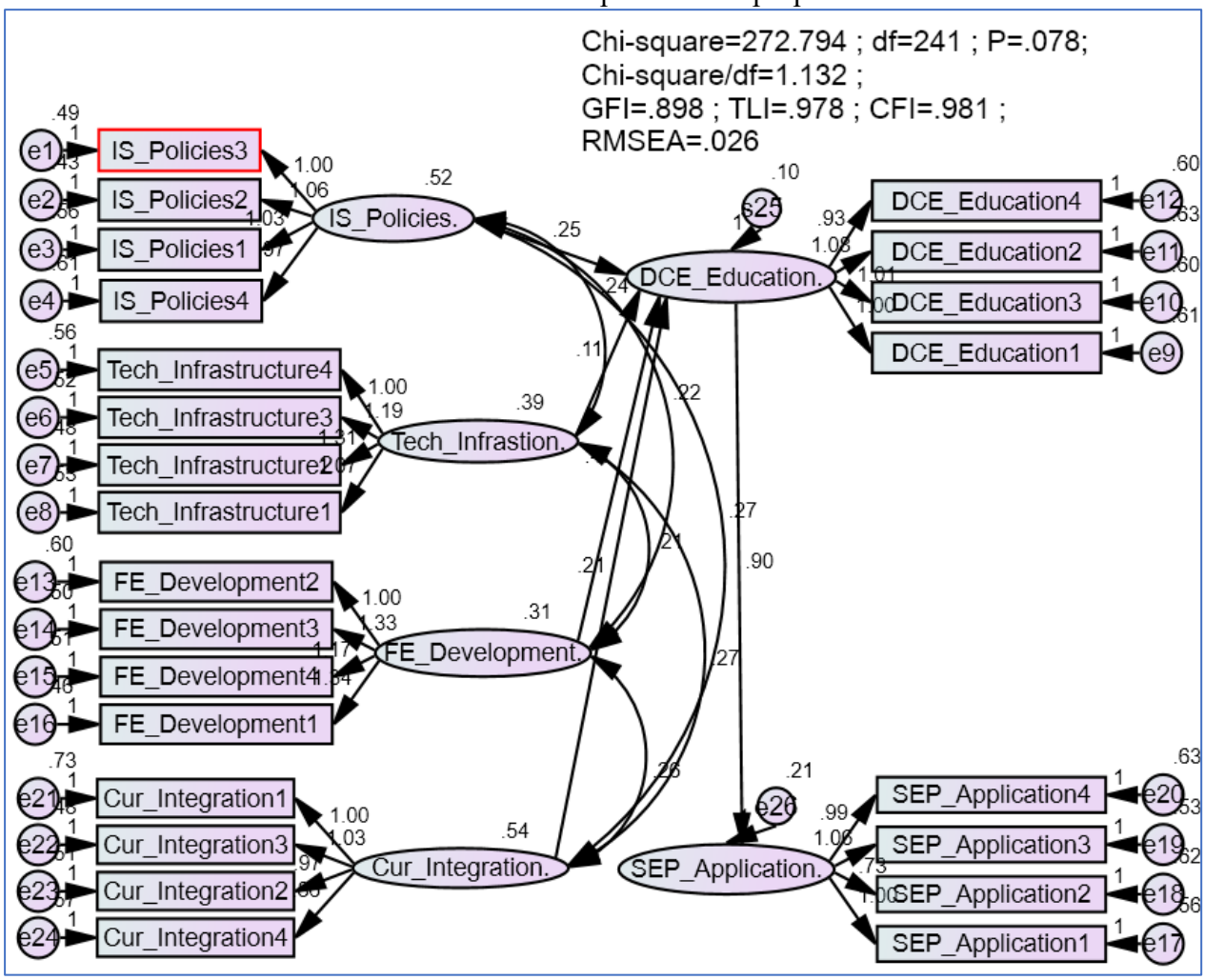


Figure 1 SEM analysis results

Hypothesis test

As evidenced in Table 4, the research results exemplify the systematic relationship that influence the socio-technical development of digital competencies in economics education. The hypotheses postulations' acceptance or rejection rested on the validity and practical effect proposed by a statistically relevant efficacy. We accepted H1 regarding curriculum integration practice based on a positive relevant relationship ($\beta = 0.210$, $p < 0.05$)

of digital competencies when curriculum involves the integration of digital strategies and tools. This indicates an essential step in designing a comprehensive curriculum that provides a framework for developing the cognitive and technical competencies required of students to help alleviate the disparate negative context and promote digital skills and technologies.

Similarly, H2 addressing faculty fluency and digital competencies; was accepted based on a strong H2 relationship ($\beta = 0.469$, $p < 0.001$) that was clearly positive. This, an indicator of the positive and essential contribution of faculty who embrace fluency with digital educational pedagogies and technologies, reconfirms the important role ongoing professional development is to cultivate educator practice that includes digital competencies and skills development.

Furthermore, we have established enough reliability to accept H4 with a predictive preferred correlation ($\beta = 0.257$, $p < 0.01$) association related to institutional support policies and its role in the credibility of digital competency. The findings are indicative of some recommended institutional types of frameworks and policies that support digital competency development in economics education. These shared associations are not unlike policy makers. and suggest that an institutional platform that supports a pragmatic attitude in education with policies very much should emphasize building the necessary capacity through structure and policy to support authentic digital training.

Finally, we provide additional evidence for H5 as clearly support was provided ($\beta = 0.902$, $p < 0.001$) confirming that digital competencies assumed relevance. The practical application of theories of economics was facilitated through student-enhanced participation and engagement confirmed from the hypothetical closeness of the association. This relationship illustrates how digital competencies can potentially impact student engagement with their learning process and prepare students to work in sectors influenced by the economy.

On the other hand, Hypothesis 3 (H3) was not supported, as the relationship between technological infrastructure and students' digital competencies was statistically non-significant ($\beta = 0.108$, $p > 0.05$). This finding suggests that mere access to technology is insufficient for cultivating meaningful digital competencies in economics education. One possible explanation lies in the distinction between access and quality: the measurement focused on general infrastructure availability (e.g., internet access, software, hardware) but may not have fully captured usability, relevance, or integration quality within pedagogical practice.

Furthermore, this result highlights a pedagogical gap—suggesting that without appropriate instructional strategies, training, and curriculum alignment, the presence of technological tools does not automatically translate into skill acquisition or effective digital engagement. This supports prior studies (e.g., Bingimlas, 2009; McKnight et al., 2016) that stress the importance of teacher mediation and digital pedagogy in leveraging infrastructure for educational outcomes. It may also reflect disparities in how technology is implemented across institutions, pointing to systemic inconsistencies in digital integration practices.

Thus, the lack of significance does not necessarily imply that infrastructure is unimportant, but rather that its impact is conditional—dependent on factors such as digital fluency of instructors, supportive institutional policies, and pedagogical design. Future research should consider disaggregating technological variables (e.g., quality, accessibility, usage intensity) and explore interaction effects between infrastructure and other contextual variables to better

understand their combined influence on digital competency development. This relationship highlights key considerations in understanding students' digital competency development in economics education, curriculum integration strategy, student technology integration along with feeding faculty capacity through professional development, institutional support while heating against too much emphasis on technology alone, without a pedagogical and strategic framework.

Discussion

The research greatly enhances our understanding of the manner and way that digital competencies are developed in economics education, and in the context of the rapid technological and economic changes in Vietnam (Quispe Mera et al., 2023), warn the authors. The research is filling an important gap in the current literature on the construction of the constituent elements of digital competency development. It demonstrates the foundational components that allow for digital competencies to be embedded in higher education.

The findings from this investigation clearly indicate that numerous determinates are significant to promote digital competencies within students, where it is essential that digital tools and methodologies are integrated into the economics curricula. The group establishes that economics curricula must be redesigned in a way that reflects the current digital demands (Sarva et al., 2023). In addition, student development of digital literacy is influenced by how important faculty use of digital tools. Therefore, it is advisable to continue to invest significantly to maximize the development of faculty members' digital capacity, and this should be influencing the students' best way forward (Tran et al., 2023), the group also argues.

An important yet under-discussed finding of this study is that technological infrastructure was not a significant predictor of students' digital competencies. While infrastructure—such as access to internet, digital libraries, and specialized software—has been emphasized in prior research (e.g., Warschauer & Matuchniak, 2010) as a key enabler of digital learning, our results suggest that infrastructure alone may function more as a "hygiene factor"—necessary for digital competency development, but insufficient on its own to generate meaningful learning outcomes.

This discrepancy may be explained by the difference between availability and effective use. While most institutions may now possess baseline digital infrastructure, without intentional pedagogical integration, supportive training, and faculty engagement, these tools may remain underutilized. In this sense, digital infrastructure may create the conditions for learning, but the actual competencies emerge only through structured engagement, curriculum design, and guided use.

The implication for policymakers and institutions is clear: future investments should not focus solely on expanding technological infrastructure, but rather on ensuring that such investments are accompanied by capacity-building for educators, curriculum integration strategies, and institutional support mechanisms. Without these complementary efforts, infrastructure risks becoming an underleveraged asset, rather than a transformative educational driver (Hoai et al., 2023). Finally, supportive institutions and policy environments will allow digital competency development to become a reality, reinforcing the need for an educational environment that offers a socializing incentive (Pham, 2023).

The research team draws the conclusion that this study points to major implications for

educators, curriculum designers, faculty development coordinators, and education policymakers, all of whom are called on to consciously pivot educational resources more decidedly toward digital education. Ongoing efforts are warranted wherever all participant educators create a plan for enriching educational programs with digital resources, which will require ongoing professional development at the educator's higher educational institution in the area of digital - technology for educators and the development of a wider cultural of digital creativity across the campus community.

Although this work is promising, the authors noted that the findings cannot be generalized; the study centered on Vietnamese factors. In addition, due to the study's cross-sectional design, the data depicted a snapshot of student attitudes and digital literacy at a specific time, rather than tracking their progress (Romero-Tena et al., 2020).

Further work could include longitudinal studies in various educational contexts, inquiring into students' attitudes toward online learning and going into further detail on how or whether we can improve the design of particular online learning technologies efficacy in enhancing educational outcomes (Ferrer et al,2022). Other avenues for future research could also seek to understand how students' socio-demographic characteristics may mediate the influence of digital competencies on their academic performance.

Conclusion

Concepts like digital literacy and digital competencies illustrate the "significance of technology in teaching and learning today." There is an urgent need to create a broader set of digital competencies – beyond computer literacy – for educators, students, and future generations. Based on this need, the study explored the factors related to building digital competencies in economics education. Therefore, a gap in the literature existed to which little work existed in regard to how to effectively use digital strategies and tools in higher education (Nguyen, 2023).

The research presented four essential purposes to investigate the impact of curriculum integration, faculty expertise, technology infrastructure and institutional support on the various elements - animated between education and digital use through competencies competencies - responsible for developing digital competencies in higher education (Tran et al., 2023; Hoai et al., 2023). As a result, the research made a contribution to this literature through a detailed analysis of what elements contribute to develop digital competencies in economics education at universities in Vietnam. Findings supported curriculum integration and faculty expertise that are important to encourage the digital development of educators, but these are only a necessary, but insufficient, condition for a high emphasis to be placed on the technology infrastructure. Furthermore, institutional support and policy were also important when looking at the effect on economists' digital competencies (Chuc & Anh, 2023). Therefore, these findings have implications for educators who need to implement a systematic approach inclusive of curriculum design, faculty development and an institutional framework that facilitates the development of educators' digital competencies.

This study has several limitations. First, the use of a cross-sectional design limits the ability to draw causal inferences. Second, data were collected through self-reported questionnaires, which may be subject to social desirability bias or recall inaccuracies. Third, while the study involved several leading Vietnamese economics universities, the findings may not be generalizable to all higher education contexts.

Future research could address these limitations by employing mixed-methods approaches, incorporating interviews or classroom observations to triangulate survey findings. Longitudinal studies would help track changes in digital competencies over time. In addition, cross-cultural comparisons could provide insights into how institutional and cultural contexts influence the development and use of digital competencies in economics education. These directions would enhance understanding and support more robust, evidence-based policy and curriculum design.

Declarations

Acknowledgments: The authors would like to express their gratitude to the lecturers who participated in the survey
Funding: This article is a research product from the project titled "Training Human Resources in Economics in Higher Education Institutions in Vietnam to Meet the Digital Economy Requirements" sponsored by Foreign Trade University. The study code for this project is B2024-NTH-01
Ethics Statements: All procedures were performed in compliance with relevant guidelines and regulations.
Conflict of Interest: The authors declare no conflicts of interest regarding the publication of this article.
Informed Consent: All survey participants provided informed consent. Participation was voluntary, and respondents were assured of the confidentiality of their responses.
Data availability: The data supporting this study's findings are available from the corresponding author upon reasonable request. Due to confidentiality agreements with participants, certain restrictions may apply.

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